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Welcome to First Principles of Computer Vision: 3D Reconstruction - Multiple Viewpoints

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Pre-Course Survey

Week 1: Introduction to First Principles of Computer Vision

Course Syllabus

Welcome to First Principles of Computer Vision: 3D Reconstruction - Multiple Viewpoints

This course focuses on the recovery of the 3D structure of a scene from images taken from different viewpoints. We start by first building a comprehensive geometric model of a camera and then develop a method for finding (calibrating) the internal and external parameters of the camera model. Then, we show how two such calibrated cameras, whose relative positions and orientations are known, can be used to recover the 3D structure of the scene. This is what we refer to as simple binocular stereo. Next, we tackle the problem of uncalibrated stereo where the relative positions and orientations of the two cameras are unknown. Interestingly, just from the two images taken by the cameras, we can both determine the relative positions and orientations of the cameras and then use this information to estimate the 3D structure of the scene.

Next, we focus on the problem of dynamic scenes. Given two images of a scene that includes moving objects, we show how the motion of each point in the image can be computed. This apparent motion of points in the image is called optical flow. Optical flow estimation allows us to track scene points over a video sequence. Next, we consider the video of a scene shot using a moving camera, where the motion of the camera is unknown. We present structure from motion that takes as input tracked features in such a video and determines not only the 3D structure of the scene but also how the camera moves with respect to the scene. The methods we develop in the course are widely used in object modeling, 3D site modeling, robotics, autonomous navigation, virtual reality and augmented reality.

Course Prerequisites

Learners are required to know the fundamentals of linear algebra and the fundamentals of calculus. No prior knowledge of imaging or computer vision is assumed. The knowledge of any programming language would enable the learners to understand how the methods described in the course can be implemented in software. However, the knowledge of a programming language is not required as the course does not have programming assignments.

Learning Objectives

- 3D reconstruction of scenes using images taken from multiple viewpoints is a fundamental problem in computer vision. It has direct applications in object modeling, 3D site modeling, robotics, autonomous navigation, virtual reality, and augmented reality. This course is designed to achieve the following learning goals:
- Develop a comprehensive model of a camera and learn how to calibrate a camera by estimating its parameters
 - Develop a simple stereo system that uses two cameras of known configuration to estimate the 3D structure of a scene
 - Learn how to perform uncalibrated stereo, where 3D scene structure can be computed even when the relative positions and orientations of the cameras are unknown
 - Develop optical flow algorithms for estimating the motion of points in a video sequence
 - Design an algorithm for recovering both the structure of the scene and the motion of the camera from a video

Course Schedule

This is a graduate-level course, with an estimated 4-6 hours of study time per week.

Grading Policy

- Quizzes count towards 100% of the grade. You will be given 2 attempts for each quiz every 8 hours.
- **A passing grade for the course is 70% or higher.**

Course Outline

Week 1: Introduction

- 1.1 Overview of Introduction
- 1.2 What is Computer Vision?
- 1.3 What is Vision Used For?
- 1.4 How Do Humans Do It?
- 1.5 Topics Covered
- 1.6 About the Lecture Series
- 1.7 References and Credits

Week 2: Camera Calibration

- 2.1 Overview of Camera Calibration
- 2.2 Linear Camera Model
- 2.3 Camera Calibration
- 2.4 Intrinsic and Extrinsic Matrices
- 2.5 Simple Stereo

Week 3: Uncalibrated Stereo

- 3.1 Overview of Uncalibrated Stereo
- 3.2 Problem of Uncalibrated Stereo
- 3.3 Epipolar Geometry
- 3.4 Stereo Vision in Nature
- 3.5 Estimating Fundamental Matrix
- 3.6 Finding Correspondences
- 3.7 Computing Depth

Week 4: Optical Flow

- 4.1 Overview of Optical Flow
- 4.2 Motion Field and Optical Flow
- 4.3 Optical Flow Constraint Equation
- 4.4 Lucas-Kanade Method
- 4.5 Coarse-to-Fine Flow Estimation
- 4.6 Application of Optical Flow

Week 5: Structure from Motion

- 5.1 Overview of Structure from Motion
- 5.2 Structure from Motion Problem
- 5.3 Observation Matrix
- 5.4 Rank of Observation Matrix
- 5.5 Tomasi-Kanade Factorization

Suggested Readings

Textbooks

- Computer Vision: Algorithms and Applications, Szeliski, R., Springer
- Computer Vision: A Modern Approach, Forsyth, D and Ponce, J., Prentice Hall
- Robot Vision, Horn, B. K. P., MIT Press
- A Guided Tour of Computer Vision, Nalwa, V., Addison-Wesley
- Digital Image Processing, González, R and Woods, R., Prentice Hall
- Optics, Hecht, E., Addison-Wesley
- Eye and Brain, Gregory, R., Princeton University Press
- Animal Eyes, Land, M. and Nilsson, D., Oxford University Press

Papers

In addition to the above textbooks, the course also presents concepts that are described in published research papers. The papers are listed at the end of the last video clip of each topic –camera calibration, uncalibrated stereo, optical flow and structure from motion.

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